Experiment -6

Python Program To Vacuum Cleaner Problem

Code:

```
def vacuum_cleaner():
    rooms = {'A': 'Dirty', 'B': 'Dirty'}
    location = 'A'

    print("Initial State:", rooms)

    for room in rooms:
        location = room
        print("Vacuum at Room", location)

        if rooms[location] == 'Dirty':
            print("Cleaning Room", location)
            rooms[location] = 'Clean'
        else:
            print("Room already clean")

    print("Final State:", rooms)

vacuum_cleaner()
```

```python
def vacuum_cleaner():
    rooms = {'A': 'Dirty', 'B': 'Dirty'}
    location = 'A'

    print("Initial State:", rooms)

    for room in rooms:
        location = room
        print("Vacuum at Room", location)

        if rooms[location] == 'Dirty':
            print("Cleaning Room", location)
            rooms[location] = 'Clean'
        else:
            print("Room already clean")

    print("Final State:", rooms)

vacuum_cleaner()
```

```
Initial State: {'A': 'Dirty', 'B': 'Dirty'}
Vacuum at Room A
Cleaning Room A
Vacuum at Room B
Cleaning Room B
Final State: {'A': 'Clean', 'B': 'Clean'}


...Program finished with exit code 0
Press ENTER to exit console.
```